

## Aluminum Electrolytic Capacitors Axial Standard

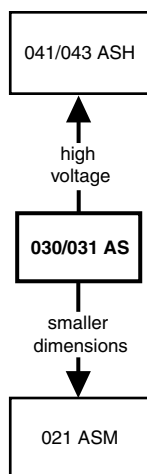
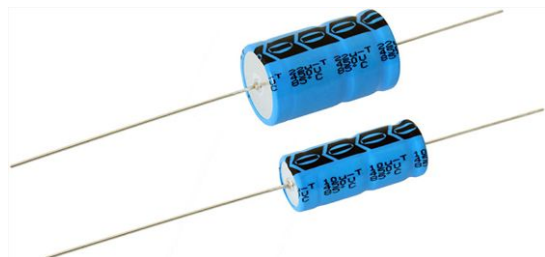


Fig. 1


**RoHS**  
COMPLIANT

### FEATURES

- Polarized aluminum electrolytic capacitors, non-solid electrolyte
- Axial leads, cylindrical aluminum case, insulated with a blue sleeve
- Taped version available for automatic insertion
- Charge and discharge proof
- Useful life: 3000 h at 85 °C
- Standard dimensions
- Material categorization: for definitions of compliance please see [www.vishay.com/doc?99912](http://www.vishay.com/doc?99912)

### APPLICATIONS

- General purpose and industrial, automotive, telecommunication, audio-video
- Coupling, decoupling, timing, smoothing, filtering, buffering in SMPS
- Boards with restricted mounting height, vibration, and shock resistant

### MARKING

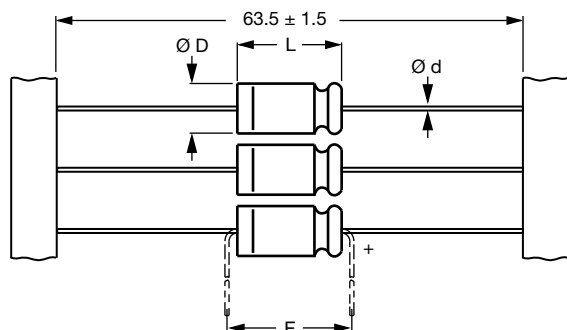
The capacitors are marked (where possible) with the following information:

- Rated capacitance (in  $\mu\text{F}$ )
- Tolerance on rated capacitance, code letter in accordance with IEC 60062 (T for -10 % to +50 %)
- Rated voltage (in V)
- Date code in accordance with IEC 60062
- Code factory of origin
- Name of manufacturer
- Negative terminal identification
- Series number (030 or 031)

| QUICK REFERENCE DATA                            |                                       |
|-------------------------------------------------|---------------------------------------|
| DESCRIPTION                                     | VALUE                                 |
| Nominal case sizes ( $\varnothing$ D x L in mm) | 4.5 x 10 to 10 x 25                   |
| Rated capacitance range, $C_R$                  | 1 $\mu\text{F}$ to 1000 $\mu\text{F}$ |
| Tolerance on $C_R$                              | -10 % to +50 %                        |
| Rated voltage, $U_R$                            | 6.3 V to 100 V                        |
| Category temperature range                      | -40 °C to +85 °C                      |
| Endurance test at 85 °C                         | 2000 h                                |
| Useful life at 85 °C                            | 3000 h                                |
| Useful life at 40 °C, 1.4 x $I_R$ applied       | 80 000 h                              |
| Shelf life at 0 V, 85 °C                        | 500 h                                 |
| Based on sectional specification                | IEC 60384-4 / EN130300                |
| Climatic category IEC 60068                     | 40 / 085 / 56                         |

**SELECTION CHART FOR  $C_R$ ,  $U_R$  AND RELEVANT NOMINAL CASE SIZES ( $\varnothing D \times L$  in mm)**

| $C_R$<br>( $\mu F$ ) | $U_R$ (V) |          |          |          |          |          |          |
|----------------------|-----------|----------|----------|----------|----------|----------|----------|
|                      | 6.3       | 10       | 16       | 25       | 40       | 63       | 100      |
| 1.0                  | -         | -        | -        | -        | -        | 4.5 x 10 | 4.5 x 10 |
| 2.2                  | -         | -        | -        | -        | -        | 4.5 x 10 | 4.5 x 10 |
| 3.3                  | -         | -        | -        | -        | -        | 4.5 x 10 | 4.5 x 10 |
| 4.7                  | -         | -        | -        | -        | -        | 4.5 x 10 | 6 x 10   |
| 6.8                  | -         | -        | -        | -        | -        | 4.5 x 10 | 6 x 10   |
| 10                   | -         | -        | -        | 4.5 x 10 | 4.5 x 10 | 6 x 10   | 8 x 11   |
|                      | -         | -        | -        | -        | -        | -        | 6.5 x 18 |
| 15                   | -         | -        | -        | -        | 4.5 x 10 | 6 x 10   | -        |
| 22                   | -         | -        | -        | 4.5 x 10 | 6 x 10   | 8 x 11   | 8 x 18   |
|                      | -         | -        | -        | -        | -        | 6.5 x 18 | -        |
| 33                   | -         | -        | 4.5 x 10 | -        | 6 x 10   | -        | 10 x 18  |
| 47                   | -         | 4.5 x 10 | -        | 6 x 10   | 8 x 11   | 8 x 18   | 10 x 25  |
|                      | -         | -        | -        | -        | 6.5 x 18 | -        | -        |
| 68                   | 4.5 x 10  | -        | 6 x 10   | -        | -        | 10 x 18  | -        |
| 100                  | -         | 6 x 10   | -        | 8 x 11   | 8 x 18   | 10 x 25  | -        |
|                      | -         | -        | -        | 6.5 x 18 | -        | -        | -        |
| 150                  | 6 x 10    | -        | 8 x 11   | 8 x 18   | 10 x 18  | -        | -        |
|                      | -         | -        | 6.5 x 18 | -        | -        | -        | -        |
| 220                  | -         | 8 x 11   | 8 x 18   | 10 x 18  | 10 x 25  | -        | -        |
|                      | -         | 6.5 x 18 | -        | -        | -        | -        | -        |
| 330                  | -         | 8 x 18   | 10 x 18  | 10 x 25  | -        | -        | -        |
| 470                  | 8 x 18    | 10 x 18  | 10 x 25  | -        | -        | -        | -        |
| 680                  | 10 x 18   | 10 x 25  | -        | -        | -        | -        | -        |
| 1000                 | 10 x 25   | -        | -        | -        | -        | -        | -        |

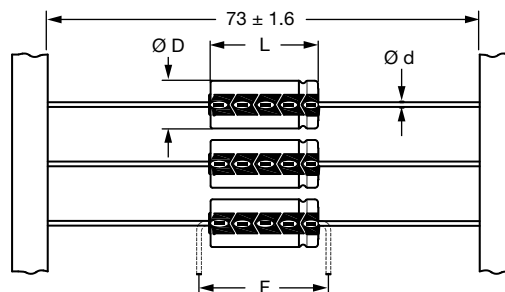
**DIMENSIONS in millimeters AND AVAILABLE FORMS**


**Form BR:** Taped on reel

**Form BA:** Taped in box (ammopack)

Case  $\varnothing D \times L = 4.5 \text{ mm} \times 10 \text{ mm}$  to  $8 \text{ mm} \times 11 \text{ mm}$

Fig. 2 - Forms BA and BR



**Form BR:** Taped on reel

Case  $\varnothing D \times L = 6.5 \text{ mm} \times 18 \text{ mm}$  to  $15 \text{ mm} \times 30 \text{ mm}$

**Form BA:** Taped in box (ammopack)

Case  $\varnothing D \times L = 6.5 \text{ mm} \times 18 \text{ mm}$  to  $10 \text{ mm} \times 25 \text{ mm}$

Fig. 3 - Forms BA and BR

Table 1

| <b>AXIAL; DIMENSIONS in millimeters, MASS AND PACKAGING QUANTITIES</b> |           |                      |                        |            |            |             |                      |         |
|------------------------------------------------------------------------|-----------|----------------------|------------------------|------------|------------|-------------|----------------------|---------|
| NOMINAL CASE SIZE<br>$\varnothing D \times L$                          | CASE CODE | AXIAL FORM BA AND BR |                        |            |            | MASS<br>(g) | PACKAGING QUANTITIES |         |
|                                                                        |           | $\varnothing d$      | $\varnothing D_{max.}$ | $L_{max.}$ | $F_{min.}$ |             | FORM BA              | FORM BR |
| 4.5 x 10                                                               | 2         | 0.6                  | 5.0                    | 10.5       | 15         | ≈ 0.5       | 1000                 | 3000    |
| 6 x 10                                                                 | 3         | 0.6                  | 6.3                    | 10.5       | 15         | ≈ 0.7       | 1000                 | 1000    |
| 8 x 11                                                                 | 5a        | 0.6                  | 8.5                    | 11.5       | 15         | ≈ 1.1       | 500                  | 500     |
| 6.5 x 18                                                               | 4         | 0.8                  | 6.9                    | 18.5       | 25         | ≈ 1.3       | 1000                 | 1000    |
| 8 x 18                                                                 | 5         | 0.8                  | 8.5                    | 18.5       | 25         | ≈ 1.7       | 500                  | 500     |
| 10 x 18                                                                | 6         | 0.8                  | 10.5                   | 18.5       | 25         | ≈ 2.5       | 500                  | 500     |
| 10 x 25                                                                | 7         | 0.8                  | 10.5                   | 25.0       | 30         | ≈ 3.3       | 500                  | 500     |

**Note**

- Detailed tape dimensions see section "PACKAGING"



| ELECTRICAL DATA |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| SYMBOL          | DESCRIPTION                                                                              |
| $C_R$           | Rated capacitance at 100 Hz, tolerance -10 % to +50 %                                    |
| $I_R$           | Rated RMS ripple current at 100 Hz, 85 °C                                                |
| $I_{L1}$        | Max. leakage current after 1 min at $U_R$                                                |
| $I_{L5}$        | Max. leakage current after 5 min at $U_R$                                                |
| $\tan \delta$   | Max. dissipation factor at 100 Hz                                                        |
| ESR             | Equivalent series resistance at 100 Hz (calculated from $\tan \delta_{\max}$ and $C_R$ ) |
| Z               | Max. impedance at 10 kHz                                                                 |

**Note**

- Unless otherwise specified, all electrical values in Table 2 apply at  $T_{\text{amb}} = 20\text{ °C}$ ,  $P = 86\text{ kPa}$  to  $106\text{ kPa}$ ,  $RH = 45\text{ %}$  to  $75\text{ %}$ .

Table 2

| ELECTRICAL DATA AND ORDERING INFORMATION |                                      |                                                          |              |                                  |                                        |                                        |                         |                               |                             |                             |                         |
|------------------------------------------|--------------------------------------|----------------------------------------------------------|--------------|----------------------------------|----------------------------------------|----------------------------------------|-------------------------|-------------------------------|-----------------------------|-----------------------------|-------------------------|
| $U_R$<br>(V)                             | $C_R$<br>100 Hz<br>( $\mu\text{F}$ ) | NOMINAL<br>CASE SIZE<br>$\varnothing D \times L$<br>(mm) | CASE<br>CODE | $I_R$<br>100 Hz<br>85 °C<br>(mA) | $I_{L1}$<br>1 min<br>( $\mu\text{A}$ ) | $I_{L5}$<br>5 min<br>( $\mu\text{A}$ ) | $\tan \delta$<br>100 Hz | ESR<br>100 Hz<br>( $\Omega$ ) | Z<br>10 kHz<br>( $\Omega$ ) | ORDERING CODE<br>MAL2.....  |                         |
|                                          |                                      |                                                          |              |                                  |                                        |                                        |                         |                               |                             | TAPED ON<br>REEL<br>FORM BR | TAPED IN BOX<br>FORM BA |
| 6.3                                      | 68                                   | 4.5 x 10                                                 | 2            | 75                               | 22                                     | 5.9                                    | 0.25                    | 5.86                          | 2.90                        | 03023689E3                  | 03033689E3              |
|                                          | 150                                  | 6 x 10                                                   | 3            | 120                              | 10                                     | 6.9                                    | 0.25                    | 2.66                          | 1.30                        | 03023151E3                  | 03033151E3              |
|                                          | 470                                  | 8 x 18                                                   | 5            | 330                              | 22                                     | 11                                     | 0.25                    | 0.85                          | 0.43                        | 03123471E3                  | 03133471E3              |
|                                          | 680                                  | 10 x 18                                                  | 6            | 430                              | 30                                     | 14                                     | 0.25                    | 0.59                          | 0.29                        | 03123681E3                  | 03133681E3              |
|                                          | 1000                                 | 10 x 25                                                  | 7            | 560                              | 42                                     | 18                                     | 0.25                    | 0.40                          | 0.20                        | 03123102E3                  | 03133102E3              |
| 10                                       | 47                                   | 4.5 x 10                                                 | 2            | 70                               | 24                                     | 5.9                                    | 0.20                    | 6.78                          | 3.40                        | 03024479E3                  | 03034479E3              |
|                                          | 100                                  | 6 x 10                                                   | 3            | 110                              | 10                                     | 7.0                                    | 0.20                    | 3.19                          | 1.60                        | 03024101E3                  | 03034101E3              |
|                                          | 220                                  | 8 x 11                                                   | 5a           | 210                              | 18                                     | 9.4                                    | 0.20                    | 1.45                          | 0.73                        | 03024221E3                  | 03034221E3              |
|                                          | 220                                  | 6.5 x 18                                                 | 4            | 210                              | 18                                     | 9.4                                    | 0.20                    | 1.45                          | 0.73                        | 03124221E3                  | 03134221E3              |
|                                          | 330                                  | 8 x 18                                                   | 5            | 310                              | 24                                     | 12                                     | 0.20                    | 0.97                          | 0.48                        | 03124331E3                  | 03134331E3              |
|                                          | 470                                  | 10 x 18                                                  | 6            | 410                              | 33                                     | 14                                     | 0.20                    | 0.68                          | 0.34                        | 03124471E3                  | 03134471E3              |
|                                          | 680                                  | 10 x 25                                                  | 7            | 510                              | 45                                     | 19                                     | 0.20                    | 0.47                          | 0.24                        | 03124681E3                  | 03134681E3              |
| 16                                       | 33                                   | 4.5 x 10                                                 | 2            | 65                               | 27                                     | 6.1                                    | 0.16                    | 7.72                          | 3.60                        | 03025339E3                  | 03035339E3              |
|                                          | 68                                   | 6 x 10                                                   | 3            | 110                              | 11                                     | 7.2                                    | 0.16                    | 3.75                          | 1.80                        | 03025689E3                  | 03035689E3              |
|                                          | 150                                  | 8 x 11                                                   | 5a           | 200                              | 19                                     | 9.8                                    | 0.16                    | 1.70                          | 0.80                        | 03025151E3                  | 03035151E3              |
|                                          | 150                                  | 6.5 x 18                                                 | 4            | 200                              | 19                                     | 9.8                                    | 0.16                    | 1.70                          | 0.80                        | 03125151E3                  | 03135151E3              |
|                                          | 220                                  | 8 x 18                                                   | 5            | 270                              | 26                                     | 12                                     | 0.16                    | 1.16                          | 0.55                        | 03125221E3                  | 03135221E3              |
|                                          | 330                                  | 10 x 18                                                  | 6            | 410                              | 36                                     | 16                                     | 0.16                    | 0.78                          | 0.36                        | 03125331E3                  | 03135331E3              |
|                                          | 470                                  | 10 x 25                                                  | 7            | 480                              | 49                                     | 20                                     | 0.16                    | 0.55                          | 0.26                        | 03125471E3                  | 03135471E3              |
| 25                                       | 10                                   | 4.5 x 10                                                 | 2            | 50                               | 13                                     | 5.5                                    | 0.14                    | 22.3                          | 9.00                        | 03026109E3                  | 03036109E3              |
|                                          | 22                                   | 4.5 x 10                                                 | 2            | 60                               | 28                                     | 6.1                                    | 0.14                    | 10.2                          | 4.10                        | 03026229E3                  | 03036229E3              |
|                                          | 47                                   | 6 x 10                                                   | 3            | 100                              | 12                                     | 7.4                                    | 0.14                    | 4.80                          | 1.90                        | 03026479E3                  | 03036479E3              |
|                                          | 100                                  | 8 x 11                                                   | 5a           | 160                              | 19                                     | 10                                     | 0.14                    | 2.23                          | 0.90                        | 03026101E3                  | 03036101E3              |
|                                          | 100                                  | 6.5 x 18                                                 | 4            | 160                              | 19                                     | 10                                     | 0.14                    | 2.23                          | 0.90                        | 03126101E3                  | 03136101E3              |
|                                          | 150                                  | 8 x 18                                                   | 5            | 240                              | 27                                     | 13                                     | 0.14                    | 1.49                          | 0.60                        | 03126151E3                  | 03136151E3              |
|                                          | 220                                  | 10 x 18                                                  | 6            | 350                              | 37                                     | 16                                     | 0.14                    | 1.02                          | 0.41                        | 03126221E3                  | 03136221E3              |
|                                          | 330                                  | 10 x 25                                                  | 7            | 460                              | 54                                     | 22                                     | 0.14                    | 0.68                          | 0.27                        | 03126331E3                  | 03136331E3              |

**ORDERING EXAMPLE**

Electrolytic capacitor 031 series

330  $\mu\text{F}$  / 10 V; -10 % / +50 %Nominal case size:  $\varnothing 8\text{ mm} \times 18\text{ mm}$ ; form BA

Ordering code: MAL203134331E3



| <b>ELECTRICAL DATA AND ORDERING INFORMATION</b> |                               |                                                          |              |                                  |                                 |                                 |                         |                               |                               |                             |                         |
|-------------------------------------------------|-------------------------------|----------------------------------------------------------|--------------|----------------------------------|---------------------------------|---------------------------------|-------------------------|-------------------------------|-------------------------------|-----------------------------|-------------------------|
| $U_R$<br>(V)                                    | $C_R$<br>100 Hz<br>( $\mu$ F) | NOMINAL<br>CASE SIZE<br>$\varnothing D \times L$<br>(mm) | CASE<br>CODE | $I_R$<br>100 Hz<br>85 °C<br>(mA) | $I_{L1}$<br>1 min<br>( $\mu$ A) | $I_{L5}$<br>5 min<br>( $\mu$ A) | $\tan \delta$<br>100 Hz | ESR<br>100 Hz<br>( $\Omega$ ) | $Z$<br>10 kHz<br>( $\Omega$ ) | ORDERING CODE<br>MAL2.....  |                         |
|                                                 |                               |                                                          |              |                                  |                                 |                                 |                         |                               |                               | TAPED ON<br>REEL<br>FORM BR | TAPED IN BOX<br>FORM BA |
| 40                                              | 10                            | 4.5 x 10                                                 | 2            | 50                               | 20                              | 5.8                             | 0.11                    | 17.6                          | 7.00                          | 03027109E3                  | 03037109E3              |
|                                                 | 15                            | 4.5 x 10                                                 | 2            | 55                               | 30                              | 6.2                             | 0.11                    | 11.7                          | 4.70                          | 03027159E3                  | 03037159E3              |
|                                                 | 22                            | 6 x 10                                                   | 3            | 75                               | 9                               | 6.8                             | 0.11                    | 8.00                          | 3.20                          | 03027229E3                  | 03037229E3              |
|                                                 | 33                            | 6 x 10                                                   | 3            | 95                               | 12                              | 7.7                             | 0.11                    | 5.31                          | 2.10                          | 03027339E3                  | 03037339E3              |
|                                                 | 47                            | 8 x 11                                                   | 5a           | 150                              | 16                              | 8.8                             | 0.11                    | 3.73                          | 1.50                          | 03027479E3                  | 03037479E3              |
|                                                 | 47                            | 6.5 x 18                                                 | 4            | 150                              | 16                              | 8.8                             | 0.11                    | 3.73                          | 1.50                          | 03127479E3                  | 03137479E3              |
|                                                 | 100                           | 8 x 18                                                   | 5            | 220                              | 28                              | 13                              | 0.11                    | 1.75                          | 0.70                          | 03127101E3                  | 03137101E3              |
|                                                 | 150                           | 10 x 18                                                  | 6            | 300                              | 40                              | 17                              | 0.11                    | 1.17                          | 0.47                          | 03127151E3                  | 03137151E3              |
|                                                 | 220                           | 10 x 25                                                  | 7            | 430                              | 57                              | 23                              | 0.11                    | 0.80                          | 0.32                          | 03127221E3                  | 03137221E3              |
| 63                                              | 1.0                           | 4.5 x 10                                                 | 2            | 13                               | 5                               | 5.1                             | 0.09                    | 143                           | 55.0                          | 03028108E3                  | 03038108E3              |
|                                                 | 2.2                           | 4.5 x 10                                                 | 2            | 25                               | 7                               | 5.3                             | 0.09                    | 65.2                          | 25.0                          | 03028228E3                  | 03038228E3              |
|                                                 | 3.3                           | 4.5 x 10                                                 | 2            | 35                               | 11                              | 5.4                             | 0.09                    | 46.5                          | 17.0                          | 03028338E3                  | 03038338E3              |
|                                                 | 4.7                           | 4.5 x 10                                                 | 2            | 40                               | 15                              | 5.6                             | 0.09                    | 30.5                          | 12.0                          | 03028478E3                  | 03038478E3              |
|                                                 | 6.8                           | 4.5 x 10                                                 | 2            | 46                               | 22                              | 5.9                             | 0.09                    | 21.1                          | 8.10                          | 03028688E3                  | 03038688E3              |
|                                                 | 10                            | 6 x 10                                                   | 3            | 70                               | 7                               | 6.3                             | 0.08                    | 12.8                          | 5.50                          | 03028109E3                  | 03038109E3              |
|                                                 | 15                            | 6 x 10                                                   | 3            | 79                               | 10                              | 6.9                             | 0.08                    | 8.50                          | 3.70                          | 03028159E3                  | 03038159E3              |
|                                                 | 22                            | 8 x 11                                                   | 5a           | 110                              | 13                              | 7.8                             | 0.08                    | 5.79                          | 2.50                          | 03028229E3                  | 03038229E3              |
|                                                 | 22                            | 6.5 x 18                                                 | 4            | 110                              | 13                              | 7.8                             | 0.08                    | 5.79                          | 2.50                          | 03128229E3                  | 03138229E3              |
|                                                 | 47                            | 8 x 18                                                   | 5            | 190                              | 22                              | 11                              | 0.08                    | 2.71                          | 1.20                          | 03128479E3                  | 03138479E3              |
|                                                 | 68                            | 10 x 18                                                  | 6            | 250                              | 30                              | 14                              | 0.08                    | 1.88                          | 0.81                          | 03128689E3                  | 03138689E3              |
|                                                 | 100                           | 10 x 25                                                  | 7            | 300                              | 42                              | 18                              | 0.08                    | 1.28                          | 0.55                          | 03128101E3                  | 03138101E3              |
| 100                                             | 1.0                           | 4.5 x 10                                                 | 2            | 20                               | 5                               | 4.6                             | 0.08                    | 128                           | 45.0                          | 03029108E3                  | 03039108E3              |
|                                                 | 2.2                           | 4.5 x 10                                                 | 2            | 30                               | 11                              | 5.3                             | 0.08                    | 57.9                          | 21.0                          | 03029228E3                  | 03039228E3              |
|                                                 | 3.3                           | 4.5 x 10                                                 | 2            | 40                               | 17                              | 6.0                             | 0.08                    | 38.6                          | 14.0                          | 03029338E3                  | 03039338E3              |
|                                                 | 4.7                           | 6 x 10                                                   | 3            | 50                               | 13                              | 6.8                             | 0.07                    | 23.7                          | 9.60                          | 03029478E3                  | 03039478E3              |
|                                                 | 6.8                           | 6 x 10                                                   | 3            | 70                               | 18                              | 8.0                             | 0.07                    | 16.4                          | 6.60                          | 03029688E3                  | 03039688E3              |
|                                                 | 10                            | 8 x 11                                                   | 5a           | 90                               | 24                              | 10                              | 0.07                    | 11.2                          | 4.50                          | 03029109E3                  | 03039109E3              |
|                                                 | 10                            | 6.5 x 18                                                 | 4            | 90                               | 24                              | 10                              | 0.07                    | 11.2                          | 4.50                          | 03129109E3                  | 03139109E3              |
|                                                 | 22                            | 8 x 18                                                   | 5            | 120                              | 48                              | 18                              | 0.07                    | 5.07                          | 2.10                          | 03129229E3                  | 03139229E3              |
|                                                 | 33                            | 10 x 18                                                  | 6            | 200                              | 70                              | 24                              | 0.07                    | 3.38                          | 1.40                          | 03129339E3                  | 03139339E3              |
|                                                 | 47                            | 10 x 25                                                  | 7            | 260                              | 98                              | 33                              | 0.07                    | 2.37                          | 0.96                          | 03129479E3                  | 03139479E3              |

| ADDITIONAL ELECTRICAL DATA         |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                               |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PARAMETER                          | CONDITIONS                                                                                                                                                                                                                              | VALUE                                                                                                                                                                                                                                                                                                         |
| <b>Voltage</b>                     |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                               |
| Surge voltage                      |                                                                                                                                                                                                                                         | $U_s \leq 1.15 \times U_R$                                                                                                                                                                                                                                                                                    |
| Reverse voltage                    |                                                                                                                                                                                                                                         | $U_{rev} \leq 1 \text{ V}$                                                                                                                                                                                                                                                                                    |
| <b>Current</b>                     |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                               |
| Leakage current                    | After 1 min at $U_R$ :<br>Case $\varnothing D \times L = 4.5 \text{ mm} \times 10 \text{ mm}$<br>Case $\varnothing D \times L = 6 \text{ mm} \times 10 \text{ mm}$ to $10 \text{ mm} \times 25 \text{ mm}$<br><br>$U_R = 100 \text{ V}$ | $I_{L1} \leq 0.05 C_R \times U_R$ or $5 \mu\text{A}$ , whichever is greater<br>$I_{L1}$ for $CV \leq 1000$ : $\leq 0.01 C_R \times U_R$ or $1 \mu\text{A}$ , whichever is greater<br>$I_{L1}$ for $CV > 1000$ : $\leq 0.006 C_R \times U_R + 4 \mu\text{A}$<br>$I_{L1} = 0.02 C_R \times U_R + 4 \mu\text{A}$ |
|                                    | After 5 min:<br>$U_R = 6.3 \text{ V}$ to $63 \text{ V}$<br>$U_R = 100 \text{ V}$                                                                                                                                                        | $I_{L5} \leq 0.002 C_R \times U_R + 5 \mu\text{A}$<br>$I_{L5} \leq 0.006 C_R \times U_R + 4 \mu\text{A}$                                                                                                                                                                                                      |
| <b>Inductance</b>                  |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                               |
| Equivalent series inductance (ESL) | Case $\varnothing D \times L$ mm:                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                               |
|                                    | 4.5 x 10                                                                                                                                                                                                                                | typ. 10 nH                                                                                                                                                                                                                                                                                                    |
|                                    | 6 x 10                                                                                                                                                                                                                                  | typ. 22 nH                                                                                                                                                                                                                                                                                                    |
|                                    | 8 x 11                                                                                                                                                                                                                                  | typ. 85 nH                                                                                                                                                                                                                                                                                                    |
|                                    | 6.5 x 18                                                                                                                                                                                                                                | typ. 25 nH                                                                                                                                                                                                                                                                                                    |
|                                    | 8 x 18                                                                                                                                                                                                                                  | typ. 40 nH                                                                                                                                                                                                                                                                                                    |
|                                    | 10 x 18                                                                                                                                                                                                                                 | typ. 61 nH                                                                                                                                                                                                                                                                                                    |
|                                    | 10 x 25                                                                                                                                                                                                                                 | typ. 38 nH                                                                                                                                                                                                                                                                                                    |
| <b>Resistance</b>                  |                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                               |
| Equivalent series resistance (ESR) | Calculated from $\tan \delta_{max}$ and $C_R$ (see table 2)                                                                                                                                                                             | $ESR = \tan \delta / 2\pi f C_R$                                                                                                                                                                                                                                                                              |

## CAPACITANCE (C)

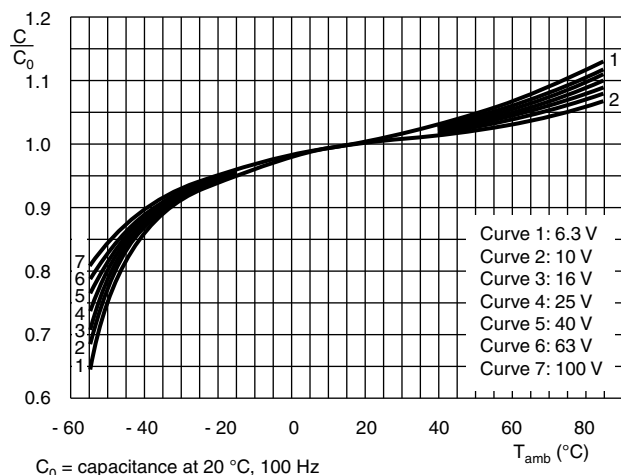


Fig. 4 - Typical multiplier of capacitance as a function of ambient temperature

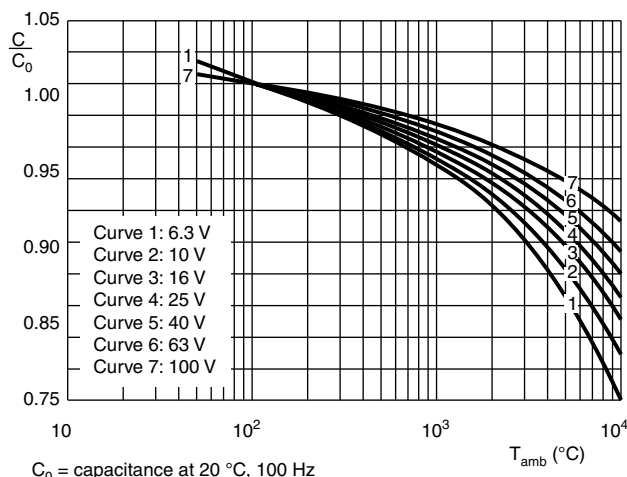


Fig. 5 - Typical multiplier of capacitance as a function of frequency

# EQUIVALENT SERIES RESISTANCE (ESR)

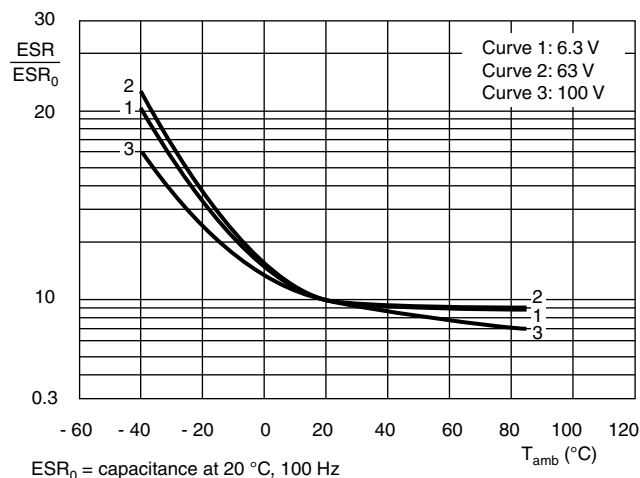


Fig. 6 - Typical multiplier of ESR as a function of ambient temperature

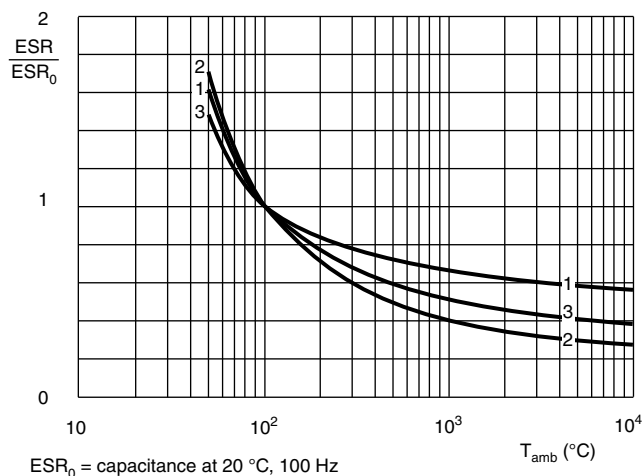


Fig. 7 - Typical multiplier of ESR as a function of frequency

# IMPEDANCE (Z)

Table 3

| IMPEDANCE VS. CAPACITANCE VALUES AT 10 kHz |                             |        |        |        |       |       |       |
|--------------------------------------------|-----------------------------|--------|--------|--------|-------|-------|-------|
| T <sub>amb</sub>                           | Z x C <sub>R</sub> (Ω x μF) |        |        |        |       |       |       |
|                                            | 6.3 V                       | 10 V   | 16 V   | 25 V   | 40 V  | 63 V  | 100 V |
| +20 °C                                     | ≤ 200                       | ≤ 160  | ≤ 120  | ≤ 90   | ≤ 70  | ≤ 55  | ≤ 45  |
| -25 °C                                     | ≤ 1200                      | ≤ 750  | ≤ 560  | ≤ 400  | ≤ 300 | ≤ 180 | ≤ 130 |
| -40 °C                                     | ≤ 3200                      | ≤ 2000 | ≤ 1500 | ≤ 1100 | ≤ 900 | ≤ 500 | ≤ 350 |

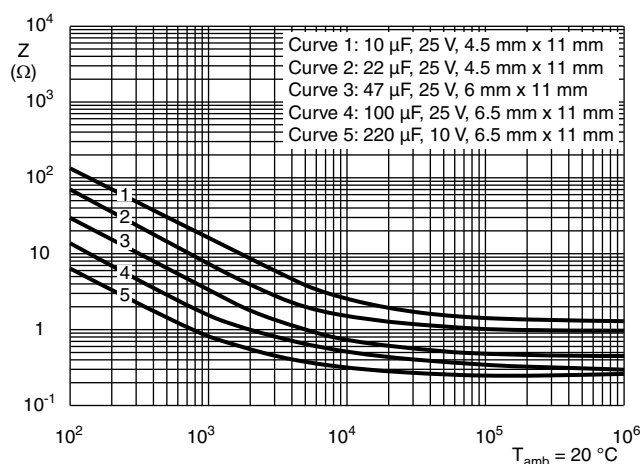


Fig. 8 - Typical impedance as a function of frequency

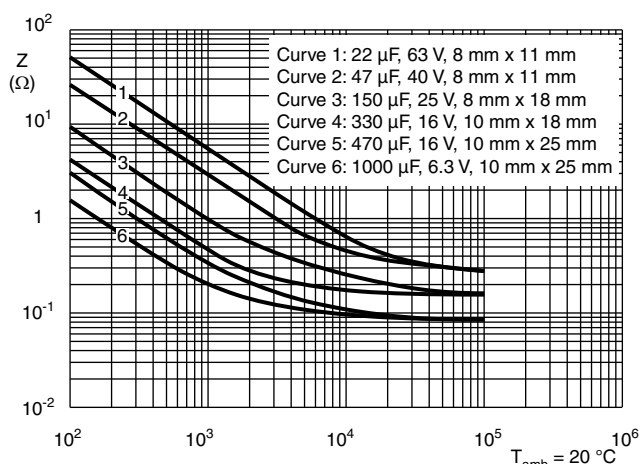


Fig. 9 - Typical impedance as a function of frequency

# RIPPLE CURRENT AND USEFUL LIFE

Table 4

| ENDURANCE TEST DURATION AND USEFUL LIFE |                          |
|-----------------------------------------|--------------------------|
| ENDURANCE AT 85 °C (h)                  | USEFUL LIFE AT 85 °C (h) |
| 2000                                    | 3000                     |

## Note

- Multiplier of useful life code: CCC205

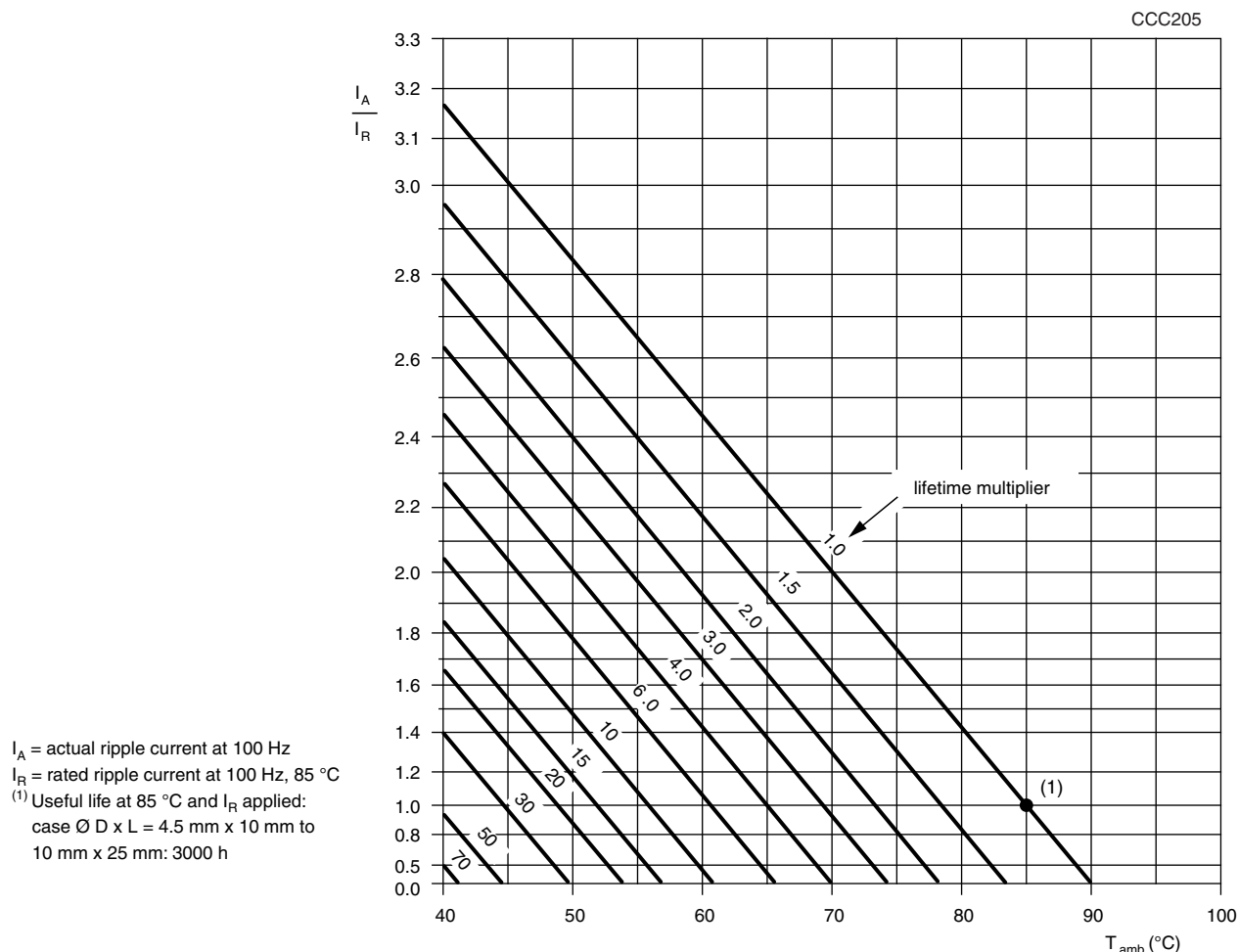


Fig. 10 - Multiplier of useful life as a function of ambient temperature and ripple current load

Table 5

| MULTIPLIER OF RIPPLE CURRENT ( $I_R$ ) AS A FUNCTION OF FREQUENCY |                  |      |      |      |      |          |
|-------------------------------------------------------------------|------------------|------|------|------|------|----------|
| $U_R$<br>(V)                                                      | FREQUENCY (Hz)   |      |      |      |      |          |
|                                                                   | 50               | 100  | 300  | 1000 | 3000 | ≥ 10 000 |
|                                                                   | $I_R$ MULTIPLIER |      |      |      |      |          |
| 6.3                                                               | 0.95             | 1.00 | 1.07 | 1.12 | 1.15 | 1.20     |
| 10                                                                | 0.95             | 1.00 | 1.07 | 1.12 | 1.15 | 1.20     |
| 16                                                                | 0.90             | 1.00 | 1.12 | 1.20 | 1.25 | 1.30     |
| 25                                                                | 0.90             | 1.00 | 1.12 | 1.20 | 1.25 | 1.30     |
| 40                                                                | 0.85             | 1.00 | 1.20 | 1.30 | 1.35 | 1.40     |
| 63                                                                | 0.85             | 1.00 | 1.20 | 1.30 | 1.35 | 1.40     |
| 100                                                               | 0.85             | 1.00 | 1.20 | 1.30 | 1.35 | 1.40     |



Table 6

| TEST PROCEDURES AND REQUIREMENTS               |                                           |                                                                                                                                                       |                                                                                                                                                                                                                                      |
|------------------------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEST                                           |                                           | PROCEDURE<br>(quick reference)                                                                                                                        | REQUIREMENTS                                                                                                                                                                                                                         |
| NAME OF TEST                                   | REFERENCE                                 |                                                                                                                                                       |                                                                                                                                                                                                                                      |
| Case Ø D x L = 4.5 mm x 10 mm to 10 mm x 25 mm |                                           |                                                                                                                                                       |                                                                                                                                                                                                                                      |
| Endurance                                      | IEC 384-4 /<br>EN130300<br>subclause 4.13 | T <sub>amb</sub> = 85 °C; U <sub>R</sub> applied;<br>2000 h                                                                                           | U <sub>R</sub> ≤ 6.3 V; ΔC/C: +15 % / -30 %<br>U <sub>R</sub> > 6.3 V; ΔC/C: ± 15 %<br>tan δ ≤ 1.3 x spec. limit<br>Z ≤ 2 x spec. limit<br>I <sub>L</sub> ≤ spec. limit                                                              |
| Useful life                                    | CECC 30301<br>subclause 1.8.1             | T <sub>amb</sub> = 85 °C; U <sub>R</sub> and I <sub>R</sub> applied;<br>3000 h                                                                        | U <sub>R</sub> ≤ 6.3 V; ΔC/C: +45 % / -50 %<br>U <sub>R</sub> > 6.3 V; ΔC/C: ± 45 %<br>tan δ ≤ 3 x spec. limit<br>Z ≤ 3 x spec. limit<br>I <sub>L</sub> ≤ spec. limit<br>no short or open circuit<br>total failure percentage: ≤ 1 % |
| Shelf life<br>(storage at<br>high temperature) | IEC 384-4 /<br>EN130300<br>subclause 4.17 | T <sub>amb</sub> = 85 °C; no voltage applied;<br>500 h<br><br>After test: U <sub>R</sub> to be applied for 30 min,<br>24 h to 48 h before measurement | ΔC/C, tan δ, Z:<br>for requirements<br>see “Endurance test” above<br>I <sub>L</sub> ≤ 2 x spec. limit                                                                                                                                |

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